

Model Feature	Purpose	Source	CPRD/HES/lit analysis required	WP1 information source	Trial data - variable	Expert elicitation Q
Route to diagnosis	Model structure	Literature, WP1 dataset and HES (A&E, OP)			Limitation: limited amount and access to data	
A – Repeated Consultations	Repeated GP consultations scheduled before condition is met – either referral, emergency diagnosis or diagnosis through inpatient or outpatient of another condition	CPRD, WP1	Data: Patient ID GP consultations: Date of consultations First consultation Last consultation Censoring – if GP consultation occurred Method: Kaplan Meier estimator for time-to-event analysis for time to next GP consultation, taking censoring into account Output: Mean time to return GP consultation Time-to-event distributions	Information: Consultation frequency analysis using a ML method to pinpoint time where the number of consultations increased likely due to cancer. SA also used. Only split for burden, not A/D Action: Has A/D been separately looked at? Source: Diagnostics timelines paper draft Consultation frequency results		Does anxiety and/or depression impact the chance of attendance of GP appointments? If so how? What do you expect the impact of the SPOCC intervention (communication tool) to be on the length of a GP appointment?
B1 – 2WW referral rate WITHOUT intervention	Probability of patient receiving a 2WW referral at GP appointment and time to	CPRD, WP1, lit	Data: Patient ID GP appointment date Referral date National data source for referral rate Method:	Information: 38.0% of A/D patients TWW referral route diagnosis (41.2% non A/D) Action: Have other referral routes been		

	referral without intervention		Calculate proportion of patients given 2WW referral after each GP appointment. Compare to national referral data source. Kaplan Meier estimator for time to referral from GP appointment. Output: Probability of 2WW referral from GP appointment Mean time to referral from GP appointment	addressed (e.g. non-2WW)? Source: Taken from route to diagnosis data in NCRAS – anxiety and depression paper		
B1 – 2WW referral rate WITH intervention	Probability of patient receiving a 2WW referral at GP appointment and time to referral with intervention	CPRD, WP1, lit, trial data	Data: Patient ID GP appointment date GP referral date Intervention completed Method: Take probability of 2WW referral and time to referral from GP appointment as above. Adjust for trial data. If trial data is insufficient: Test as a parameter to vary in threshold analysis. Output: Probability of 2WW referral from GP appointment Mean time to referral from GP appointment		Action: Will there be sufficient data on the patients' referrals after the intervention is administered?	What do you expect the 2WW referral rate (%) for patients with pre-existing anxiety and/or depression to be following the introduction of the intervention?

B2 – non-2WW referral rate WITHOUT intervention	Probability of patient receiving a non-2WW referral at GP appointment and time to referral without intervention	CPRD, WP1, lit	Data: Patient ID GP appointment date GP referral date National data source for referral rate Method: Calculate proportion of patients given non-2WW referral after each GP appointment. Compare to national referral data source. Kaplan Meier estimator for time to referral from GP appointment. Output: Probability of non-2WW referral from GP appointment Mean time to referral from GP appointment	Information: 38.0% of A/D patients TWW referral route diagnosis (41.2% non A/D) Action: Have other referral routes been addressed (e.g. non-2WW)? Source: Taken from route to diagnosis data in NCRAS – anxiety and depression paper		
B2 – non-2WW referral rate WITH intervention	Probability of patient receiving a non-2WW referral at GP appointment and time to referral with intervention	CPRD, WP1, lit, trial data	Data: Patient ID GP appointment date GP referral date Intervention completed Method: Take probability of non-2WW referral and time to referral from GP appointment as above. Adjust for trial data. If trial data is insufficient:		Action: Will there be sufficient data on the patients' referrals after the intervention is administered?	

			Test as a parameter to vary in threshold analysis. Output: Probability of non-2WW referral from GP appointment Mean time to referral from GP appointment			
C – DNA referral with return to GP appointment	Patients who do not attend either their 2WW or non-2WW referral appointment	CPRD, HES	Data: Patient ID Referral date GP consultation dates Diagnosis date CPRD referral route Scheduled referral date HES OP – DNA status Method: Proportion of patients recorded as DNA given a referral. Link CPRD referral to HES OP data – match based on GP referral date and referral appointment date. Kaplan Meier estimator for time-to-event analysis for time to next GP appointment after not attending referral appointment Limitation: may have high levels of missing data – no referral appointment may be interpreted as missed referral. Be aware of cancelled/rescheduled referrals			Does anxiety and/or depression impact the chance of attendance of referrals related to cancer diagnosis? If so how?

			Output: Probability of DNA given 2WW referral Mean time to attending GP appointment after DNA			
D1 – Emergency only	Patients who get diagnosed in emergency without a referral issued from GP	CPRD, WP1, HES	Data: Patient ID GP appointment date Cancer symptom Date of A&E visit Date of diagnosis Method: Proportion of emergency diagnosis given no prior referrals - Use linked HES and CPRD to remove proportion of patients who had previous events in diagnosis pathway. Kaplan Meier estimator time to diagnosis. Output: Probability of emergency diagnosis given no prior referrals issued from GP. Mean time to diagnosis.	Information: 23.4% of A/D patients diagnosed after emergency presentation (19.8% non A/D) Action: Use HES to remove routes with prior referrals Source: Taken from route to diagnosis data in NCRAS – anxiety and depression paper		What do you expect the likelihood (%) for patients with pre-existing anxiety and/or depression of an immediate emergency referral to be given the symptoms identified through the intervention?
D2 – outpatient/inpatient only	Patients who get diagnosed in out/inpatient for another condition without a	CPRD, HES	Data: Patient ID GP appointment date Primary diagnosis (not cancer) Admission/discharge/ appointment date			

	referral issued from GP		Diagnosis date Method: Proportion of diagnoses in out/inpatient given no prior referrals - Use linked HES and CPRD to remove proportion of patients who had previous events in diagnosis pathway. Kaplan Meier estimator time to diagnosis. Output: Probability of diagnosis in in/outpatient without a referral issued from GP. Mean time to diagnosis.			
E1 – No diagnosis from 2WW Consider removing if limited occurrences	Patients who attend their 2WW referral appointment but do not receive a diagnosis	CPRD, HES	Data: Patient ID Referral type Referral date No cancer diagnosis in HES code following referral Method: Proportion of patients who do not receive a diagnosis given a 2WW referral. Kaplan Meier estimator time to no diagnosis. Output: Probability not receiving cancer diagnosis for patients who	Action: Check that ‘no cancer diagnosis’ is recorded in available HES data		

			attend their 2WW referral appointment. Mean time to no diagnosis.			
E2 – Diagnosis from 2WW	Patients who receive a diagnosis from an urgent referral appointment	CPRD, HES, WP1	Data: Patient ID Diagnosis date CPRD diagnosis route HES route – date and type of referral Method: Proportion of patients who receive a diagnosis following 2WW referral. Link to HES. Remove patients who DNA or emergency Kaplan Meier estimator for time to diagnosis from referral. Output: Probability of diagnosis given 2WW. Mean time to diagnosis from referral.	Information: 38.0% of A/D patients diagnosed after 2WW referral (41.2% non A/D) Action: Confirm all possible patient routes included Source: Taken from route to diagnosis data in NCRAS – anxiety and depression paper		
F1 – No diagnosis from non-2WW Consider removing if limited occurrences	Patients who attend their non-2WW referral appointment but do not receive a diagnosis	CPRD, HES	Data: Patient ID Referral type Referral date No cancer diagnosis in HES code following referral Method:	Action: Check that ‘no cancer diagnosis’ is recorded in available HES data		

			Proportion of patients who do not receive a diagnosis given a non-2WW referral. Kaplan Meier estimator time to no diagnosis. Output: Probability not receiving cancer diagnosis for patients who attend their non-2WW referral appointment. Mean time to no diagnosis.			
F2 – Diagnosis from non-2WW	Patients who receive a diagnosis from a non-urgent referral	CPRD, HES	Data: Patient ID Diagnosis date CPRD diagnosis route HES route – date and type of referral Method: Proportion of patients who receive a diagnosis following non-2WW referral. Link to HES. Remove patients who DNA or emergency Kaplan Meier estimator for time to diagnosis from referral. Output: Probability of diagnosis given 2WW. Mean time to diagnosis from referral.			

G – Consultation after no initial diagnosis Consider removing if limited occurrences	Patients who return to GP after not receiving their diagnosis in the current referral	CPRD, HES	Data: Patient ID CPRD diagnosis route HES route – date and type of referral GP appointment date GP cancer symptoms Method: Proportion of patients who go back to their GP after not receiving a diagnosis from a referral Kaplan Meier estimator for time to GP appointment from not receiving diagnosis. Output: Probability of GP appointment given negative diagnosis results Mean time to return GP appointment.			
H1 – 2WW -> emergency diagnosis	Patients with prior 2WW referral but emergency diagnosis received before attendance	CPRD, HES	Data: Patient ID CPRD diagnosis route HES route Referral dates Date of A&E visit Date of diagnosis Method: Use linked HES and CPRD to determine proportion of patients who received a 2WW referral but diagnosed in	Information: 23.4% of A/D patients diagnosed after emergency presentation (19.8% non A/D) Includes true emergency patients Action: Split into different diagnosis routes Source:		

			emergency before referral appointment diagnosis. Kaplan Meier estimator for time to diagnosis from GP consultation. Output: Probability of emergency given 2WW referral issued but not attended. Mean time to diagnosis.	Taken from route to diagnosis data in NCRAS – Anxiety and depression paper		
H2 – 2WW -> out/inpatient diagnosis Consider removing if limited occurrences	Patients with prior 2WW referral but outpatient or inpatient (from another condition) diagnosis received before attendance	CPRD, HES	Data: Patient ID Primary diagnosis (not cancer) CPRD diagnosis route HES route Referral dates Admission/discharge/ appointment date Diagnosis date Method: Use linked HES and CPRD to determine proportion of patients who received a 2WW referral but diagnosed in out/inpatient before referral appointment diagnosis. Kaplan Meier estimator for time to diagnosis from GP consultation. Output:			

			Probability of out/inpatient diagnosis given 2WW referral issued but not attended. Mean time to diagnosis.			
I1 – non-2WW -> emergency diagnosis	Patients with prior non-2WW referral but emergency diagnosis received before attendance	CPRD, HES	Data: Patient ID CPRD diagnosis route HES route Referral dates Date of A&E visit Date of diagnosis Method: Use linked HES and CPRD to determine proportion of patients who received a non-2WW referral but diagnosed in emergency before referral appointment diagnosis. Kaplan Meier estimator for time to diagnosis from GP consultation. Output: Probability of emergency given non-2WW referral issued but not attended. Mean time to diagnosis.	Information: 23.4% of A/D patients diagnosed after emergency presentation (19.8% non A/D) Includes true emergency patients Action: Split into different diagnosis routes Source: Taken from route to diagnosis data in NCRAS – Anxiety and depression paper		
I2 – non-2WW -> out/inpatient diagnosis Consider removing if	Patients with prior non-2WW referral but outpatient or inpatient (from another	CPRD, HES	Data: Patient ID Primary diagnosis (not cancer) CPRD diagnosis route HES route Referral dates			

limited occurrences	condition) diagnosis received before attendance		Admission/discharge/ appointment date Diagnosis date Method: Use linked HES and CPRD to determine proportion of patients who received a non-2WW referral but diagnosed in out/inpatient before referral appointment diagnosis. Kaplan Meier estimator for time to diagnosis from GP consultation. Output: Probability of out/inpatient diagnosis given non-2WW referral issued but not attended. Mean time to diagnosis.			
J1 – DNA -> emergency diagnosis Are patients with A/D more likely to get diagnosed through emergency with this type of pathway?	Patients who did not attend a referral and received an emergency diagnosis	CPRD, HES	Data: Patient ID CPRD diagnosis route HES route GP consultation dates Referral dates Date of A&E visit Date of diagnosis Method: Use linked HES and CPRD to determine proportion of patients who received a referral but DNA and diagnosed in emergency.			

			Kaplan Meier estimator for time to diagnosis from GP consultation. Output: Probability of emergency given referral issued but DNA. Mean time to diagnosis.			
J2 – DNA -> out/inpatient diagnosis Consider removing if limited occurrences	Patients who did not attend a referral and received a diagnosis as an out/inpatient for another condition	CPRD, HES	Data: Patient ID Primary diagnosis (not cancer) GP consultation dates CPRD diagnosis route HES route Referral dates Admission/discharge/ appointment date Diagnosis date Method: Use linked HES and CPRD to determine proportion of patients who received a referral but DNA and diagnosed in out/inpatient. Kaplan Meier estimator for time to diagnosis from GP consultation. Output: Probability of out/inpatient diagnosis given referral issued but DNA. Mean time to diagnosis.			

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Model Feature	Purpose	Source	CPRD/HES/lit analysis required	WP1 information source	Trial data - variable	Expert elicitation Q
Survival and Utilities						
		Data sources				
Survival	By cancer type	Literature, CPRD, WP1	Survival after diagnosis of either breast, prostate, lung or colon cancer			
	For early-stage cancers – stages 1 and 2	Literature, CPRD stage_best	Survival after diagnosis of stage 1 or 2	WP1 recorded stage diagnosis from NCRAS in two groups – early stage (0,1,2), late stage (3,4)		
	For late-stage cancers – stages 3 and 4	Literature	Survival after diagnosis of stage 3 or 4	49.2% A/D patients had advanced-stage cancer diagnosis (50.1% non A/D)		
	For emergency diagnosis	Literature	Probability of death after diagnosis through emergency referral			
	For 2WW diagnosis	Literature	Probability of death after diagnosis through 2WW referral			
	For non-2WW diagnosis	Literature	Probability of death after diagnosis through non-2WW referral			
Death – no diagnosis		Literature, life tables	Probability of death with no cancer diagnosis			
Utilities	Before cancer diagnosis	Literature	Baseline QoL before any referrals or cancer diagnosis			

	After referral issued WITH intervention	Literature	QoL after referral issued from GP appointment given symptoms have been identified through intervention			
	After referral issued WITHOUT intervention	Literature	QoL after referral issued from GP appointment			
	After cancer diagnosis	Literature	QoL after cancer diagnosis			
Disutilities	Decrease in QoL from symptoms identified in intervention	Literature	Decrease in QoL from symptoms identified in intervention			
	Decrease in QoL from cancer diagnosis - type	Literature	Decrease in QoL from cancer diagnosis depending on type – breast, prostate, lung or colon			What are the impacts of a cancer diagnosis on QoL for a patient with anxiety and/or depression?
	Decrease in QoL from cancer diagnosis – early stages	Literature	Decrease in QoL from cancer diagnosis depending on stage – type 1 or 2			
	Decrease in QoL from cancer diagnosis – late stages	Literature	Decrease in QoL from cancer diagnosis depending on stage – type 3 or 4			
Suicide risk	Before cancer diagnosis	Literature				
	After referral issued	Literature				

	After cancer diagnosis	Literature				What is your estimate of the suicide risk (%) for a patient with pre-existing anxiety and/or depression around the time of a cancer diagnosis?
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Costs and resources						
Cost area	Specific Cost	Staging dependent? Stages 1,2 vs stages 3,4	Cancer type dependent? Breast, prostate, lung or colon	Data sources	Additional notes	Expert elicitation
Primary care and referral costs	GP appointments	No	No	PSSRU	- Number of follow up appointments to reach a diagnosis - Type of appointment e.g. face to face or in person	
	2WW referral letters and appointments	No	No	NHS reference costs, PSSRU, expert elicitation		
	Non-2WW referral letters and appointments	No	No	NHS reference costs, PSSRU		

	DNA appointments	No	No	NHS reference costs, PSSRU		
Intervention costs	Additional GP appointment time	No	No	Expert elicitation, trial data, PSSRU		What is the impact of the intervention (communication tool) on the length of a GP appointment expected to be (difference from current mean appointment length)? <i>Units in minutes, allow negative as well as positive values</i>
	Material and administration	No	No	Trial data, reference costs		
	Intervention tool implementation	No	No	Expert elicitation, trial data		What resources will be required to implement the intervention across multiple practices?
	Intervention training	No	No	Trial data		
	Intervention uptake	No	No	Trial data		
Emergency and in/outpatient costs	Emergency attendance	No	No	CPRD/HES, NHS reference costs	- Cost codes associated with the emergency diagnosis of breast, prostate, lung or colon cancer	

	Additional in/outpatient costs	No	No	CPRD/HES, NHS reference costs		
Diagnosis costs	Referral diagnosis	Yes	Yes		- Breast – mammograms, ultrasounds, biopsy, MRI - Prostate – PSA, MRI, biopsy - Lung – x-ray, CT, biopsy - Colon – FIT, colonoscopy, CT	
	Emergency diagnosis	Yes	Yes			
Treatment costs	Drugs and medication	Yes	Yes	Literature, NHS guidelines, product SmPCs, eMIT, BNF	- To include drug dosages, duration and drug costs - Treatment duration and number of treatments - Breast – surgery, radiotherapy, hormone therapy, chemotherapy, targeted therapy - Prostate – surgery, surveillance, hormone therapy, radiotherapy - Lung – surgery, chemotherapy, radiotherapy, targeted therapy, immunotherapy - Colon – surgery, chemotherapy, radiotherapy	
	Surgery costs	Yes	Yes	NHS reference costs		
	Radiotherapy, chemotherapy	Yes	Yes	NHS reference costs		
	Hospitalisation	Yes	Yes	NHS reference costs		
	Outpatient visits	Yes	Yes	NHS reference costs		
	Aftercare	Yes	Yes	NHS reference costs		
	Surveillance/ monitoring	Yes	Yes	NHS reference costs		

Patient details						
Characteristic	Data sources	Notes				
Population size	Trial data, CPRD	Data used to check that the trial population is similar to the CPRD/HES population to generalise the trial data to practises across the country				
Patient ID	CPRD					
Age	Trial data, CPRD					
Gender	Trial data, CPRD					
Ethnicity	Trial data, CPRD					
Socioeconomic background	Trial data, CPRD					
Other comorbidities	Trial data, CPRD					